Lecture 28

The centre of $U(\mathfrak{g})$ and central characters

Objectives. *Central elements of $U(\mathfrak{g})$ act by scalars on highest-weight modules. We compute these scalars, find the whole centre for $\mathfrak{sl}_2$ and state Harish-Chandra's theorem. A central character depends on a highest weight λ only through the Weyl orbit of $\lambda+\rho$. We compare this orbit with the Casimir sphere of Lecture 16. The lecture ends by collecting what a highest-weight label gives.*

By the end of this lecture you will be able to:

- Prove that the quadratic Casimir acts on a highest-weight module of highest weight λ by $(\lambda, \lambda+2\rho)$, and convert this eigenvalue between normalisations of the invariant form.
- Prove $Z(U(\mathfrak{sl}_2)) = \mathbb{C}[C]$ using the PBW theorem and Verma modules.
- Compute a central character from the Harish-Chandra projection, shifted by ρ and evaluated at $\lambda+\rho$.
- Explain why central characters separate finite-dimensional simple modules but not Verma modules, and why the quadratic Casimir alone need not separate finite-dimensional simple modules.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, positive roots R_+, Weyl group W, $\rho = \frac{1}{2}\sum_{\alpha\in R_+}\alpha$ and $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$. Fix a nondegenerate invariant symmetric form $(\cdot,\cdot)$ on $\mathfrak{g}$, normalised on each simple factor so that long roots have squared length 2. The same symbol denotes the induced form on $\mathfrak{h}^*$.

Departure from the standing convention. The algebra $U(\mathfrak{g})$, Verma modules and other modules in this lecture may be infinite-dimensional. All arguments are algebraic.

28.1 Central characters

The *centre* of $U(\mathfrak{g})$ is

$$Z(U(\mathfrak{g})) = \{z \in U(\mathfrak{g}) : zu = uz \text{ for all } u \in U(\mathfrak{g})\}.$$

Write Z for short. Since $\mathfrak{g}$ generates $U(\mathfrak{g})$,

$$z \in Z \iff xz = zx \quad \text{for all } x \in \mathfrak{g}.$$

A central element acts on every module by an intertwiner. Which modules can these operators tell apart?

Root spaces are one-dimensional (Lecture 8). The orthogonality argument of Lecture 7 uses only invariance and nondegeneracy. Hence $(\cdot,\cdot)$ pairs $\mathfrak{g}_\alpha$ with $\mathfrak{g}_{-\alpha}$ and is nondegenerate on $\mathfrak{h}$. For each $\alpha \in R_+$ we can therefore choose

$$e_\alpha \in \mathfrak{g}_\alpha, \qquad f_\alpha \in \mathfrak{g}_{-\alpha}, \qquad (e_\alpha, f_\alpha) = 1.$$

Choose an orthonormal basis $u_1, \dots, u_r$ of $\mathfrak{h}$.

List R_+ as $\beta_1, \dots, \beta_N$ and write $e_j = e_{\beta_j}$, $f_j = f_{\beta_j}$. By the PBW theorem (Lecture 11) the ordered monomials

$$f^a u^b e^c = f_1^{a_1} \cdots f_N^{a_N}\, u_1^{b_1} \cdots u_r^{b_r}\, e_1^{c_1} \cdots e_N^{c_N}$$

form a basis of $U(\mathfrak{g})$. Since $\mathfrak{h}$ is abelian, $U(\mathfrak{h}) = \mathrm{Sym}(\mathfrak{h})$ is a polynomial algebra. We read $p \in U(\mathfrak{h})$ as a polynomial function on $\mathfrak{h}^*$. The value $p(\lambda)$ is obtained by the substitution

$$u \longmapsto \lambda(u) \quad \text{for } u \in \mathfrak{h}.$$

Definition 28.1 (Harish-Chandra projection)**.** The *Harish-Chandra projection* is the linear map $\mathrm{pr}\colon U(\mathfrak{g}) \to U(\mathfrak{h})$ with

$$\mathrm{pr}(f^a u^b e^c) = \begin{cases} u^b, & a = 0 \text{ and } c = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 28.2 (Action on a highest-weight vector $\cdot$ *P*)**.** *Let $z \in U(\mathfrak{g})$ satisfy $uz = zu$ for all $u \in \mathfrak{h}$. Let V be a $\mathfrak{g}$-module and $v \in V$ a vector with*

$$uv = \lambda(u)v \quad \textit{for all } u \in \mathfrak{h}, \qquad \mathfrak{n}_+ v = 0.$$

Then

$$zv = \mathrm{pr}(z)(\lambda)\, v.$$

Proof. For $u \in \mathfrak{h}$, the map $w \mapsto uw - wu$ is a derivation of $U(\mathfrak{g})$. Each factor e_j contributes $\beta_j(u)$, each f_j contributes $-\beta_j(u)$ and each u_i contributes 0. Hence, on a PBW monomial,

$$[u, f^a u^b e^c] = \Big(\sum_{j=1}^{N} (c_j - a_j)\beta_j\Big)(u)\; f^a u^b e^c.$$

Since $[u, z] = 0$ for all $u \in \mathfrak{h}$ and the monomials form a basis, z is a combination of monomials with $\sum_j (c_j - a_j)\beta_j = 0$.

If $c \neq 0$, the monomial ends with some e_j and kills v. If $c = 0$, the weight condition reads

$$\sum_{j=1}^{N} a_j \beta_j = 0.$$

If some $a_j > 0$, the left side has nonnegative simple-root coefficients, not all zero, so it is nonzero. Therefore $a = 0$. The surviving monomials u^b act on v by $u^b(\lambda)$, and their sum is $\mathrm{pr}(z)(\lambda)$. ∎

A *highest-weight module* of highest weight λ is a module $V = U(\mathfrak{g})v$ generated by a nonzero vector v as in the lemma, its *highest-weight vector*.

Proposition 28.3 (Central characters · *P*)**.** *Every $z \in Z$ acts on a highest-weight module of highest weight λ by the scalar*

$$\chi_\lambda(z) = \mathrm{pr}(z)(\lambda).$$

The map $\chi_\lambda : Z \to \mathbb{C}$ and the restriction $\mathrm{pr}|_Z : Z \to U(\mathfrak{h})$ are algebra homomorphisms.

Proof. A central z commutes with $\mathfrak{h}$, so $zv = \chi_\lambda(z)v$ by the lemma. Every vector of V has the form wv with $w \in U(\mathfrak{g})$, and

$$z(wv) = w(zv) = \chi_\lambda(z)\, wv.$$

Since $v \neq 0$, the scalar by which z acts is determined by zv. Hence zz' acts by $\chi_\lambda(z)\chi_\lambda(z')$, and χ_λ is multiplicative. For every $\lambda \in \mathfrak{h}^*$ the Verma module $M(\lambda)$ is a highest-weight module of highest weight λ. On $M(\lambda)$ this gives

$$\mathrm{pr}(zz')(\lambda) = \chi_\lambda(zz') = \chi_\lambda(z)\,\chi_\lambda(z') = \mathrm{pr}(z)(\lambda)\ \mathrm{pr}(z')(\lambda).$$

Polynomial functions that agree at every point are equal. Hence $\mathrm{pr}(zz') = \mathrm{pr}(z)\,\mathrm{pr}(z')$. ∎

The homomorphism χ_λ is the *central character*. Since $\mathrm{pr}(z)(\lambda)$ is the scalar of z on $M(\lambda)$, the map $\mathrm{pr}|_Z$ does not depend on the chosen bases.

28.2 The quadratic Casimir

This section returns to the Casimir element of Lecture 16. The scalar is the one found there. The centrality proof is new, since it uses the invariant tensor of Lecture 27. We repeat the eigenvalue calculation because it also gives $\mathrm{pr}(\Omega)$ and shows how to compute the value of a central character on the Casimir element.

The *quadratic Casimir* is the element

$$\Omega = \sum_{i=1}^{r} u_i^2 + \sum_{\alpha \in R_+} (e_\alpha f_\alpha + f_\alpha e_\alpha).$$

The basis $(a_k) = (u_i, e_\alpha, f_\alpha)$ has dual basis $(a^k) = (u_i, f_\alpha, e_\alpha)$. Thus $\Omega = \sum_k a_k a^k$, with $(a_k, a^l) = \delta_{kl}$.

The form is nondegenerate on $\mathfrak{h}$. Hence each $\lambda \in \mathfrak{h}^*$ determines a unique $t_\lambda \in \mathfrak{h}$ with

$$(t_\lambda, u) = \lambda(u) \quad \text{for all } u \in \mathfrak{h}.$$

Proposition 28.4 (Quadratic Casimir eigenvalue · *P*)**.** *The element Ω is central. On every highest-weight module of highest weight λ it acts by*

$$\chi_\lambda(\Omega) = (\lambda, \lambda + 2\rho).$$

Proof. Centrality. Apply the lemma on the invariant tensor of a form in Lecture 27 to the adjoint module $\mathfrak{g}$ and the chosen invariant form. It gives the invariant tensor $\sum_k a_k \otimes a^k$. Multiplication

$$m : \mathfrak{g} \otimes \mathfrak{g} \longrightarrow U(\mathfrak{g}), \qquad a \otimes b \longmapsto ab,$$

is equivariant for the adjoint actions, since

$$[x, ab] = [x, a]b + a[x, b].$$

Thus $[x, \Omega] = 0$ for every $x \in \mathfrak{g}$, so Ω is central.

Rewriting. The bracket $[e_\alpha, f_\alpha]$ lies in $\mathfrak{g}_0 = \mathfrak{h}$. For $u \in \mathfrak{h}$,

$$(u, [e_\alpha, f_\alpha]) = ([u, e_\alpha], f_\alpha) = \alpha(u)(e_\alpha, f_\alpha) = \alpha(u).$$

Thus $[e_\alpha, f_\alpha] = t_\alpha$, which gives the first identity below in $U(\mathfrak{g})$. The second follows from the linearity of $\lambda \mapsto t_\lambda$:

$$e_\alpha f_\alpha = f_\alpha e_\alpha + t_\alpha, \qquad \sum_{\alpha \in R_+} t_\alpha = t_{2\rho}.$$

Substituting both identities into Ω gives

$$\Omega = \sum_i u_i^2 + t_{2\rho} + 2 \sum_{\alpha \in R_+} f_\alpha e_\alpha.$$

Eigenvalue. Apply this form of Ω to a highest-weight vector v of weight λ. The terms $f_\alpha e_\alpha$ kill v, since $\mathfrak{n}_+ v = 0$. Orthonormality gives

$$t_\lambda = \sum_{i=1}^{r} \lambda(u_i)\, u_i, \qquad (\lambda, \lambda) = (t_\lambda, t_\lambda) = \sum_{i=1}^{r} \lambda(u_i)^2.$$

Hence

$$\Omega v = \Big(\sum_i \lambda(u_i)^2 + \lambda(t_{2\rho}) \Big) v = \big((\lambda, \lambda) + (\lambda, 2\rho) \big) v.$$

Here the defining condition for t_λ gives

$$\lambda(t_{2\rho}) = (t_\lambda, t_{2\rho}) = (\lambda, 2\rho).$$

By Proposition 28.3 the same scalar holds on the whole module. ∎

The proof also reads off $\mathrm{pr}(\Omega) = \sum_i u_i^2 + t_{2\rho}$, so

$$\mathrm{pr}(\Omega)(\lambda) = (\lambda, \lambda + 2\rho) = (\lambda + \rho, \lambda + \rho) - (\rho, \rho). \tag{28.1}$$

The second form depends on λ only through $\lambda + \rho$.

Normalisation. Multiplying the invariant form by $a \neq 0$ divides each vector of the dual basis by a. Hence it divides the Casimir and all its eigenvalues by a. Sheet 28.1 compares trace and Killing forms.

28.3 The centre of $U(\mathfrak{sl}_2)$

For $\mathfrak{sl}_2$ use the trace form, whose roots have squared length 2. Identify $\mathfrak{h}^*$ with $\mathbb{C}$ by $\lambda \mapsto \lambda(h)$. Then

$$\alpha = 2, \qquad \rho = 1, \qquad W = \{\pm 1\}.$$

Since $\mathrm{tr}(h^2) = 2$, the vector $u_1 = h/\sqrt{2}$ forms an orthonormal basis of $\mathfrak{h}$. Since $\mathrm{tr}(ef) = 1$, the Casimir is the element below. Its second form uses $ef = fe + h$:

$$C = \frac{1}{2}h^2 + ef + fe = \frac{1}{2}h^2 + h + 2fe.$$

In the PBW order (f, h, e), the projection keeps $\frac{1}{2}h^2 + h$. Hence

$$\mathrm{pr}(C)(\lambda) = \frac{1}{2}\lambda^2 + \lambda = \frac{1}{2}\big((\lambda + 1)^2 - 1\big).$$

Recall the Verma module $M(\lambda)$ from Lecture 12. Its highest-weight vector is v_0, and it has the basis $v_j = f^j v_0$ for $j \geq 0$. Set $v_{-1} = 0$. The action is

$$hv_j = (\lambda - 2j)v_j, \qquad fv_j = v_{j+1}, \qquad ev_j = j(\lambda - j + 1)v_{j-1}. \tag{28.2}$$

Theorem 28.5 (The rank-one centre · *P*)**.** *The centre $Z(U(\mathfrak{sl}_2))$ is the polynomial ring $\mathbb{C}[C]$. The map* pr *sends it isomorphically onto*

$$\{p \in \mathbb{C}[\lambda] : p(\lambda) = p(-\lambda - 2)\}.$$

Proof. Step 1: shape. Let $z \in Z$. The monomial $f^a h^b e^c$ has $\operatorname{ad} h$-eigenvalue $2(c-a)$. Since $[h, z] = 0$, only monomials with $a = c$ occur. Hence

$$z = \sum_{k=0}^{K} f^k \, p_k(h) \, e^k, \qquad p_0 = \operatorname{pr}(z),$$

for polynomials p_k.

Step 2: injectivity. Suppose $\operatorname{pr}(z) = 0$ but some $p_k \neq 0$. Take the least such k. Then $k \geq 1$, since $p_0 = \operatorname{pr}(z) = 0$. In $M(\lambda)$, the operator e^l kills v_k for $l > k$. The terms with $l < k$ vanish by the choice of k, so only $f^k p_k(h) e^k$ acts on v_k. Moreover $p_k(h) v_0 = p_k(\lambda) v_0$. By (28.2),

$$e^k v_k = \prod_{j=1}^{k} j(\lambda - j + 1)\, v_0 = k!\, \lambda(\lambda-1)\cdots(\lambda-k+1)\, v_0.$$

Hence

$$z v_k = k!\, \lambda(\lambda-1)\cdots(\lambda-k+1)\, p_k(\lambda)\, v_k.$$

By Proposition 28.3, z acts on $M(\lambda)$ by $\operatorname{pr}(z)(\lambda) = 0$. The polynomial $\lambda(\lambda-1)\cdots(\lambda-k+1)p_k(\lambda)$ therefore vanishes for every $\lambda \in \mathbb{C}$, so $p_k = 0$. This contradiction shows that $\operatorname{pr}|_Z$ is injective.

Step 3: symmetry. Let $m \in \mathbb{Z}_{\geq 0}$. In $M(m)$,

$$e v_{m+1} = (m+1)(m-m) v_m = 0.$$

The vector v_{m+1} has weight $-m-2$. Lemma 28.2 and Proposition 28.3 give

$$\operatorname{pr}(z)(-m-2)\, v_{m+1} = z v_{m+1} = \operatorname{pr}(z)(m)\, v_{m+1}.$$

The polynomial $\operatorname{pr}(z)(\lambda) - \operatorname{pr}(z)(-\lambda-2)$ vanishes at every $\lambda = m \in \mathbb{Z}_{\geq 0}$. It has infinitely many zeros, so it is zero.

Step 4: generation. Put $p = \operatorname{pr}(z)$ and $P(\mu) = p(\mu - 1)$. Step 3 gives

$$P(-\mu) = p(-\mu - 1) = p(-(\mu-1) - 2) = p(\mu - 1) = P(\mu).$$

An even polynomial has the form $P(\mu) = Q(\mu^2)$ for a polynomial Q. Set $z' = Q(2C+1) \in \mathbb{C}[C]$. On $M(\lambda)$,

$$2C + 1 \text{ acts by } \lambda^2 + 2\lambda + 1 = (\lambda+1)^2,$$
$$z' \text{ acts by } Q\big((\lambda+1)^2\big) = P(\lambda+1) = p(\lambda).$$

Thus $z - z' \in Z$ acts on every $M(\lambda)$ by 0, so $\operatorname{pr}(z - z') = 0$. Step 2 gives $z = z'$.

Step 5: independence and image. Let Q be a polynomial with $Q(C) = 0$. Then $Q(C)$ acts by 0 on every $M(\lambda)$:

$$Q\Big(\frac{1}{2}\lambda^2 + \lambda\Big) = 0 \quad \text{for all } \lambda \in \mathbb{C}.$$

As λ varies, $\frac{1}{2}\lambda^2 + \lambda$ takes infinitely many values, so $Q = 0$. By Step 3, $\operatorname{pr}(Z)$ consists of symmetric polynomials. The computation in Step 4 uses only the symmetry of p. Hence every symmetric p equals $\operatorname{pr}(Q(2C+1))$ for the corresponding Q. ∎

In the shifted variable $\mu = \lambda + 1$, the image consists of the even, that is W-invariant, polynomials in μ. This is the rank-one case of the next theorem.

28.4 Harish-Chandra's theorem

Define the shift $\tau : U(\mathfrak{h}) \to U(\mathfrak{h})$ by $(\tau p)(\nu) = p(\nu - \rho)$. Set $\gamma = \tau \circ \mathrm{pr}|_Z : Z \to U(\mathfrak{h})$. Then, for $z \in Z$,

$$\chi_\lambda(z) = \mathrm{pr}(z)(\lambda) = \gamma(z)(\lambda + \rho). \tag{28.3}$$

Thus a central character is computed in three steps.

(i) Project z to $\mathrm{pr}(z)$, using the PBW order $\mathfrak{n}_-, \mathfrak{h}, \mathfrak{n}_+$.

(ii) Shift the variable: $\gamma(z)(\nu) = \mathrm{pr}(z)(\nu - \rho)$.

(iii) Evaluate $\gamma(z)$ at $\nu = \lambda + \rho$.

For the Casimir, (28.1) gives

$$\gamma(\Omega)(\nu) = (\nu, \nu) - (\rho, \rho).$$

The group W acts on polynomial functions on $\mathfrak{h}^*$ by

$$(wp)(\nu) = p(w^{-1}\nu).$$

The invariant polynomials form $\mathrm{Sym}(\mathfrak{h})^W$. Define the *dot action* on weights by

$$w \cdot \lambda = w(\lambda + \rho) - \rho.$$

For $z \in Z$, the condition that $\gamma(z)$ be W-invariant is

$$\mathrm{pr}(z)(w \cdot \lambda) = \mathrm{pr}(z)(\lambda) \quad \text{for all } w \in W,\ \lambda \in \mathfrak{h}^*.$$

Theorem 28.6 (Harish-Chandra · *C*)**.** *Let $\mathfrak{g}$ be a complex semisimple Lie algebra, with Cartan subalgebra $\mathfrak{h}$, positive system R_+, Weyl group W and ρ as above. Then γ is an algebra isomorphism*

$$\gamma : Z(U(\mathfrak{g})) \xrightarrow{\ \cong\ } \mathrm{Sym}(\mathfrak{h})^W.$$

Kirillov, §8.8, proves the isomorphism. Carter, Chapter 11, is a further source. Humphreys, §23.3, proves the linkage statement of Corollary 28.7 directly. In this lecture, Theorem 28.5 proves the case $\mathfrak{g} = \mathfrak{sl}_2$.

Corollary 28.7 (Linkage · *P given Theorem 28.6*)**.** *For $\lambda, \mu \in \mathfrak{h}^*$,*

$$\chi_\lambda = \chi_\mu \iff \mu + \rho \in W(\lambda + \rho).$$

Proof. First suppose $\mu + \rho = w(\lambda + \rho)$ with $w \in W$. By Theorem 28.6, each $\gamma(z)$ is W-invariant. With (28.3) this gives $\chi_\mu(z) = \chi_\lambda(z)$ for every $z \in Z$.

Conversely, suppose $\mu + \rho \notin W(\lambda + \rho)$. Then the finite orbits $O = W(\lambda + \rho)$ and $O' = W(\mu + \rho)$ are disjoint. For each $\nu \in O$ choose $\varphi_\nu \in \mathfrak{h} = (\mathfrak{h}^*)^*$ with $\varphi_\nu(\nu' - \nu) \neq 0$ for all $\nu' \in O'$. This is possible because each $\nu' - \nu$ is nonzero, so φ_ν only has to avoid finitely many hyperplanes of $\mathfrak{h}$. Put

$$p(x) = \prod_{\nu \in O} \varphi_\nu(x - \nu), \qquad P = \prod_{w \in W} wp.$$

Then P is W-invariant, since W permutes its factors. The factor p for $w = 1$ vanishes on O, so P vanishes on O. Each factor satisfies $(wp)(\mu + \rho) = p(w^{-1}(\mu + \rho))$ with $w^{-1}(\mu + \rho) \in O'$, and p has no zero on O'. Hence $P(\mu + \rho) \neq 0$. By Theorem 28.6, $P = \gamma(z)$ for some $z \in Z$, and

$$\chi_\lambda(z) = P(\lambda + \rho) = 0 \neq P(\mu + \rho) = \chi_\mu(z).$$

∎

Theorem 28.8 (Separation · *P given Theorem 28.6*). *For $\lambda, \mu \in P_+$, $L(\lambda) \cong L(\mu)$ if and only if $\chi_\lambda = \chi_\mu$.*

Proof. Isomorphic modules have equal central scalars. Conversely, suppose $\chi_\lambda = \chi_\mu$. Corollary 28.7 gives $\mu + \rho = w(\lambda + \rho)$ for some $w \in W$. By Lecture 9, $\langle \rho, \alpha_i^\vee \rangle = 1$ for each simple root, so

$$\langle \lambda + \rho, \alpha_i^\vee \rangle = \lambda(h_i) + 1 \geq 1,$$

and similarly for μ. Thus $\lambda + \rho$ and $\mu + \rho$ are dominant integral weights in one W-orbit. Each such orbit has a unique dominant member (Lecture 9). Hence $\lambda + \rho = \mu + \rho$, and $\lambda = \mu$. ∎

Verma modules behave differently. Let $\lambda \in P_+$ and $w \in W$. Since $(w \cdot \lambda) + \rho = w(\lambda + \rho)$, Corollary 28.7 gives $\chi_{w\cdot\lambda} = \chi_\lambda$. Hence $M(w \cdot \lambda)$ has central character χ_λ.

If $w \cdot \lambda$ is dominant integral, both $w(\lambda + \rho)$ and $\lambda + \rho$ are dominant in the same orbit. The uniqueness argument in the preceding proof gives $w \cdot \lambda = \lambda$. Thus only $M(\lambda)$ among these Verma modules has a finite-dimensional simple quotient (Lecture 12, Lecture 13).

Figure 28.1 shows this for $\mathfrak{sl}_3$ and $\lambda = 0$, in Dynkin labels $(a, b) = a\omega_1 + b\omega_2$. Since $w \cdot 0 + \rho = w\rho$ and W preserves lengths, the six weights $w \cdot 0$ lie on a circle centred at $-\rho$, of radius $\|\rho\| = \sqrt{2}$. The six Verma modules $M(w \cdot 0)$ have one central character. Only the weight $(0, 0)$ is dominant, so only $M(0)$ has a finite-dimensional simple quotient.

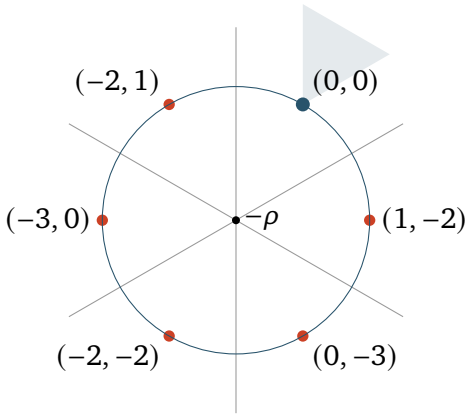


Figure 28.1: The six weights $w \cdot 0$ for $\mathfrak{sl}_3$, marked by their Dynkin labels, on the circle of radius $\sqrt{2}$ centred at $-\rho$. The grey lines are the reflecting lines of the dot action, and the shaded sector is part of the dominant chamber.

This answers the question of Section 28.1: χ_λ determines exactly the orbit $W(\lambda + \rho)$.

The Casimir sphere and the whole centre. We can now compare this answer with Lecture 16. There, the single central element Ω restricted the highest weights in the proof of the character formula to the Casimir sphere

$$S_\lambda = \{\mu \in P : \|\mu + \rho\|^2 = \|\lambda + \rho\|^2\}.$$

Equality of the whole central character restricts them to one orbit under the dot action, by Corollary 28.7. For $\lambda \in P_+$, this orbit is the index set of the character identity of Lecture 15 and Lecture 16:

$$\operatorname{ch} L(\lambda) = \sum_{w \in W} \det(w) \operatorname{ch} M(w \cdot \lambda).$$

The signs describe a subtraction of characters. They do not give a decomposition of $L(\lambda)$ into Verma submodules. In Lecture 16, dominance and the norm removed the other points

of the sphere. That argument does not use Theorem 28.6, and the comparison here is not a new proof of the character formula.

A sphere can contain more than one orbit. Exercise 16.4 gives an example. Take $\mathfrak{sl}_3$ with $\lambda = 3\omega_1$ and $\mu = 3\omega_2$. In Dynkin labels,

$$\lambda + \rho = (4,1), \qquad \mu + \rho = (1,4), \qquad \|\lambda+\rho\|^2 = \|\mu+\rho\|^2 = \frac{2}{3}(16+4+1) = 14.$$

By (28.1), Ω acts by $14 - 2 = 12$ on both $L(\lambda)$ and $L(\mu)$. Figure 28.2 shows the vectors $\lambda+\rho$ and $\mu+\rho$ on one circle. Both vectors are dominant and they differ, so they lie in different W-orbits (Lecture 9). Hence $\chi_\lambda \neq \chi_\mu$ by Corollary 28.7, and some central element separates the two modules.

We can specify one such element. As in Exercise 28.3, identify $\mathfrak{h}^*$ with $\{\nu \in \mathbb{C}^3 : \nu_1+\nu_2+\nu_3 = 0\}$, so that $W = S_3$ permutes the coordinates. Then

$$\lambda + \rho = (3,-1,-2), \qquad \mu + \rho = (2,1,-3).$$

The polynomial $p_3(\nu) = \nu_1^3 + \nu_2^3 + \nu_3^3$ is W-invariant, so Theorem 28.6 gives $z_3 \in Z$ with $\gamma(z_3) = p_3$. By (28.3),

$$\chi_\lambda(z_3) = 27 - 1 - 8 = 18, \qquad \chi_\mu(z_3) = 8 + 1 - 27 = -18.$$

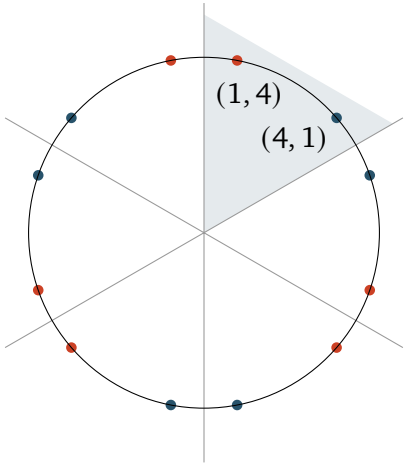


Figure 28.2: The twelve weights $\nu \in P$ of $\mathfrak{sl}_3$ with $\|\nu\|^2 = 14$, on one circle centred at 0, with colours for the two W-orbits, those of $(4,1)$ and $(1,4)$. The grey lines are the reflecting lines, and the shaded sector is part of the dominant chamber.

28.5 Worked example: $M(m)$, $M(-m-2)$ and $L(m)$

This section compares three modules with one central character. Figure 28.3 shows the case $m = 2$ of the example below. The upper row is the Verma module $M(2)$, with the coefficients of f and e. The coefficient of e from v_3 to v_2 vanishes, so the shaded tail is a submodule.

The middle row shows the images of v_0, v_1, v_2 in the quotient. These vectors do not span a submodule of $M(2)$, because the arrow f from v_2 enters the tail. The lower line shows the reflection $\lambda \mapsto -\lambda - 2$ about -1. It pairs 2 with -4, and the shifted values $\lambda + 1$ are 3 and -3.

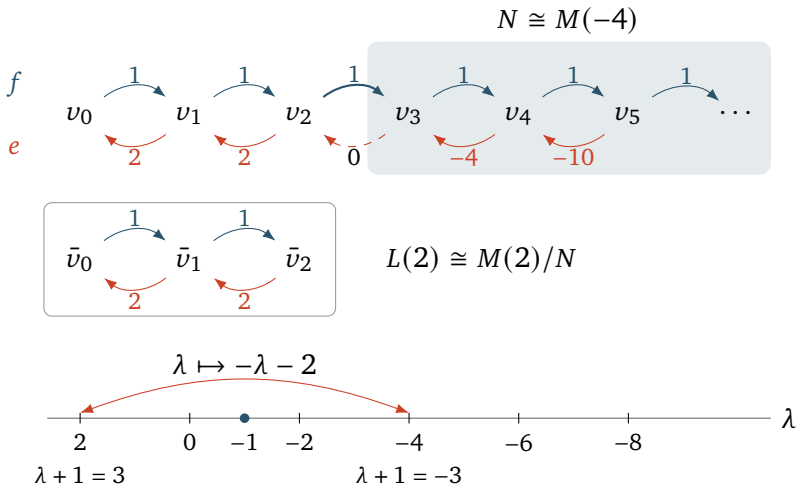


Figure 28.3: The Verma module $M(2)$ with the tail $N \cong M(-4)$ shaded, and the simple quotient $L(2)$, whose box holds the images in the quotient. The lowest line shows the reflection about -1 that pairs the two highest weights.

Example 28.9 (One central character, three modules). Fix $m \in \mathbb{Z}_{\geq 0}$. Use (28.2) and $C = \frac{1}{2}h^2 + h + 2fe$. Since $fev_j = j(\lambda - j + 1)v_j$,

$$\begin{aligned} Cv_j &= \Big(\frac{1}{2}(\lambda - 2j)^2 + (\lambda - 2j) + 2j(\lambda - j + 1)\Big)v_j \\ &= \Big(\frac{1}{2}\lambda^2 - 2j\lambda + 2j^2 + \lambda - 2j + 2j\lambda - 2j^2 + 2j\Big)v_j \\ &= \Big(\frac{1}{2}\lambda^2 + \lambda\Big)v_j. \end{aligned}$$

Hence C acts on all of $M(\lambda)$ by $\frac{1}{2}\lambda^2 + \lambda$. For $\lambda = m$ this is $\frac{1}{2}m(m+2)$. For $\lambda = -m-2$ the scalar is the same:

$$\frac{1}{2}(m+2)^2 - (m+2) = \frac{1}{2}(m^2 + 2m) = \frac{1}{2}m(m+2).$$

In $M(m)$ consider the span

$$N = \operatorname{span}\{v_j : j \geq m+1\}.$$

The subspace N is stable under h and f. It is stable under e, because $ev_j \in \mathbb{C}v_{j-1}$ and $ev_{m+1} = 0$. Let w_0 be the highest-weight vector of $M(-m-2)$. The vector v_{m+1} has weight $-m-2$ and is killed by e. By the universal property (Lecture 12) there is therefore a module map $M(-m-2) \to N$ sending $f^k w_0$ to v_{m+1+k}. This map sends a basis to a basis, so $N \cong M(-m-2)$.

The quotient $M(m)/N$ has the basis $\bar{v}_0, \dots, \bar{v}_m$ of images of $v_0, \dots, v_m$. Its action is that of V_m in Lecture 6:

$$\begin{aligned} h\bar{v}_j &= (m-2j)\bar{v}_j, & e\bar{v}_j &= j(m-j+1)\bar{v}_{j-1}, \\ f\bar{v}_j &= \bar{v}_{j+1}\ (j < m), & f\bar{v}_m &= 0. \end{aligned}$$

The case $m = 1$ recovers the defining matrices of Lecture 1. Sheet 28.1 computes the Casimir matrix on V_2.

The module V_m is simple (Lecture 6), so $M(m)/N \cong L(m)$. Hence there is an exact sequence

$$0 \longrightarrow M(-m-2) \longrightarrow M(m) \longrightarrow L(m) \longrightarrow 0.$$

The three modules compare as follows.

module	λ	$\lambda+\rho$	dimension	$\chi_\lambda(C)$
$M(m)$	m	$m+1$	infinite	$\frac{1}{2}m(m+2)$
$M(-m-2)$	$-m-2$	$-m-1$	infinite	$\frac{1}{2}m(m+2)$
$L(m)$	m	$m+1$	$m+1$	$\frac{1}{2}m(m+2)$

Since $Z(U(\mathfrak{sl}_2))=\mathbb{C}[C]$ by Theorem 28.5, every central element acts by the same scalar on all three modules. This central character determines only the orbit $\{\pm(m+1)\}$ of $\lambda+\rho$. Of the corresponding highest weights m and $-m-2$, only m is dominant integral, as in the discussion of Verma modules after Theorem 28.8.

The sequence does not split. Indeed, a splitting would give $M(m)=A\oplus B$ with $A,B\neq 0$. Both summands are then proper, so both lie in the unique maximal proper submodule (Lecture 12). Hence so does $A+B=M(m)$, a contradiction.

The singular vector v_{m+1} generates the submodule N. Weyl's theorem (Lecture 5) excludes such nonsplit extensions among finite-dimensional modules.

28.6 Synthesis and the two sequels

This section collects what the course has built. Fix $\lambda\in P_+$. By Lecture 13, the label λ names one finite-dimensional simple module $L(\lambda)$. By Weyl's theorem (Lecture 5), every finite-dimensional module is a direct sum of such simple modules. So a question about finite-dimensional modules can be asked label by label.

The table lists these questions and the tool that answers each one. Its last column gives the answers for the adjoint module of $\mathfrak{sl}_3$, the running example of Part II. Its highest weight is $\theta=\omega_1+\omega_2$. As in Lecture 19, we write 8 for this module and 10, 10^*, 27 for $L(3\omega_1)$, $L(3\omega_2)$, $L(2\theta)$.

Question	Tool	Adjoint module of $\mathfrak{sl}_3$
Dimension	Weyl dimension formula, Lecture 17	8
Character	Weyl character formula, Lecture 16; Kostant's formula, Lecture 17	$2+\sum_{\alpha\in R}e^{\alpha}$
Tensor products	Product of characters, Lecture 15; subtraction from the top weight, Lecture 17; Littlewood–Richardson coefficients in type A, Lecture 19	$8\otimes 8\cong 1\oplus 2\cdot 8\oplus 10\oplus 10^*\oplus 27$
Restriction	Interlacing in type A, Lecture 20; orthogonal, spin and symplectic cases, Lecture 26	$V_2\oplus 2V_1\oplus V_0$ over the upper-left $\mathfrak{sl}_2$
Model	Schur modules in type A, Lecture 18; harmonic Schur modules in type C, Lecture 22; tensors, Lecture 23, and spin modules, Lecture 24, in types B and D	$\mathbb{S}_{(2,1)}(\mathbb{C}^3)$, as an $\mathfrak{sl}_3$-module
Commutant	Multiplicity spaces, Lecture 27	$\dim \mathrm{End}_{\mathfrak{g}}(8\otimes 8)=8$
Central character	$\chi_\lambda(z)=\gamma(z)(\lambda+\rho)$, Section 28.4	$\chi_\theta(z)=\gamma(z)(2\rho)$, with values 6 on Ω and 0 on z_3

Several entries need a remark. The restriction names an embedding, here the upper-left block of Lecture 20. The model in the table describes the adjoint module as an $\mathfrak{sl}_3$-module, because Lecture 20 shows that the adjoint action of $\mathfrak{gl}_3$ twists $\mathbb{S}_{(2,1)}(\mathbb{C}^3)$ by $\mathbb{C}_{-\mathrm{tr}}$. In types B and D, tensors reach only the dominant weights with integer coordinates (Lecture 23), so half-integral weights need the spin modules.

The central character uses $\theta = \rho$, so $\lambda + \rho = 2\rho$. In the coordinates of Section 28.4, $2\rho = (2, 0, -2)$. Hence $\chi_\theta(\Omega) = \|2\rho\|^2 - \|\rho\|^2 = 8 - 2 = 6$ and $\chi_\theta(z_3) = p_3(2, 0, -2) = 0$.

These answers are linked. The character determines the module (Lecture 15), so it determines the dimension, the tensor products and the restriction to a named subalgebra. By Theorem 28.8, the whole central character determines the label. The single value on Ω need not, as Figure 28.2 shows for $\mathfrak{sl}_3$.

The table concerns finite-dimensional modules. Lecture 1 gave a module of an abelian algebra with an invariant line and no invariant complement. Weyl's theorem excludes this for finite-dimensional modules of a semisimple algebra. The exact sequence of Section 28.5 shows that finite dimension is a necessary hypothesis.

Two planned sequels continue from here.

Lie groups. The first sequel fixes a complex Lie group G with Lie algebra $\mathfrak{g}$. It asks which $L(\lambda)$ arise from representations of G. The answer depends on G.

Quantum groups. The sequel deforms the relations of $U(\mathfrak{g})$ by a parameter q. It asks how tensor products and complete reducibility change. It develops a parameter-dependent rule for exchanging tensor factors in place of the flip of Lecture 14, with precise conditions on q. At roots of unity, complete reducibility can fail. These are questions for the sequels; no result here depends on them.

Self-checks

(1) Starting from the highest weight of the adjoint module of $\mathfrak{sl}_3$, derive the scalar 6 by which Ω acts.

(2) Which $\mathfrak{sl}_2$ Verma modules share the central character of the trivial module? Which share that of $M(-1)$?

(3) In $\mathfrak{sl}_3$, $L(\omega_1)$ and $L(\omega_2)$ have the same Ω-eigenvalue. Does this contradict Theorem 28.8?

(4) For the module $L(2\omega_1)$ of $\mathfrak{sl}_3$, find the dimension, the restriction to the upper-left $\mathfrak{sl}_2$ and the value of Ω, as in the table of Section 28.6.

Answers. (1) The highest weight is $\lambda = \omega_1 + \omega_2 = \rho$ and $(\rho, \rho) = 2$, so $(\lambda, \lambda + 2\rho) = (\rho, 3\rho) = 6$. (2) $\lambda + 1 = \pm 1$ gives $M(0)$ and $M(-2)$. For $M(-1)$, $\lambda + 1 = 0$, so only $M(-1)$. (3) No. Both give $(\omega_i, \omega_i + 2\rho) = \frac{8}{3}$, but the theorem compares whole central characters. A cubic central element separates them (Exercise 28.3). (4) Dimension 6, restriction $V_2 \oplus V_1 \oplus V_0$ and Ω-value $\frac{26}{3} - 2 = \frac{20}{3}$. The module is $\mathbb{S}_{(2)}(\mathbb{C}^3) = \operatorname{Sym}^2 \mathbb{C}^3$.

Summary

On a highest-weight module, a central element z acts by $\chi_\lambda(z) = \gamma(z)(\lambda+\rho)$. Harish-Chandra's isomorphism makes these scalars depend exactly on the orbit $W(\lambda + \rho)$. They separate finite-dimensional simple modules. The nonsplit sequence $0 \to M(-m-2) \to M(m) \to L(m) \to 0$ above shows their limits in infinite dimension. For $\lambda \in P$, the value on Ω restricts $\lambda + \rho$ to a sphere in the real weight space. This sphere can contain several Weyl orbits.

Section 28.6 collects what a label in P_+ gives: dimension, character, tensor products, restriction, a model and a central character.

Reading

Humphreys, §§22–24; Kirillov, §8.8; Carter, Chapter 11. The isomorphism and linkage sources are specified in Section 28.4.